

1. What is Airflow?

Apache Airflow is an **open-source workflow orchestration tool** used to schedule, manage, and monitor data pipelines.

It allows you to define workflows as code using Python.

For example, a workflow might:

Extract data → Transform data → Load data → Send notification

Airflow ensures that these tasks run in the correct order and at the correct time.

Key features:

- **Scheduling** (daily, hourly, weekly, etc.)
- **Workflow orchestration** (manage dependencies between tasks)
- **Monitoring** through a web UI
- **Retries** for failed tasks
- **Logging** and alerting
- **Python-based pipeline definitions**

2. Why use Airflow instead of Azure Databricks pipelines?

It depends on the use case.

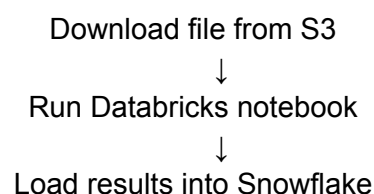
Use Azure Databricks pipelines when:

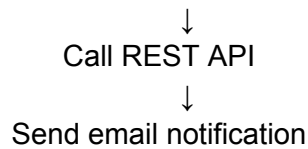
- Most of your work is inside Databricks.
- You are building ETL/ELT pipelines using Spark.
- You need Delta Live Tables (DLT).
- Your workflow mainly consists of notebooks and SQL.

Use Airflow when:

- You need to orchestrate **multiple systems**, not just Databricks.
- Your workflow involves many tools and services.

Example:



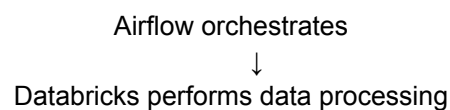


Airflow can orchestrate all these heterogeneous tasks.

Comparison

Feature	Airflow	Azure Databricks Pipelines
Open source	Yes	No
Multi-platform orchestration	Excellent	Limited
Databricks integration	Good	Native
Spark processing	Via operators	Native
Python workflow definition	Yes	Limited
Best for	Enterprise orchestration	Databricks-centric ETL

In many organizations, both are used together:



3. What is a DAG?

A **DAG (Directed Acyclic Graph)** is the core concept in Airflow.

A DAG represents a workflow as a collection of tasks and their dependencies.

DAG stands for:

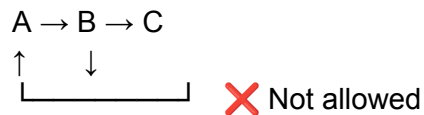
- **Directed** → Tasks have a specific execution order.
- **Acyclic** → No loops are allowed.
- **Graph** → Tasks are connected as nodes.

Example:

Task A → Task B → Task C

- Task B starts only after Task A finishes.
- Task C starts only after Task B finishes.

A DAG cannot contain cycles such as:



Simple Airflow DAG example

```
from airflow import DAG
from airflow.operators.empty import EmptyOperator
from datetime import datetime
```

```
with DAG(
    dag_id="my_first_dag",
    start_date=datetime(2026, 1, 1),
    schedule="@daily"
) as dag:
```

```
task1 = EmptyOperator(task_id="start")
task2 = EmptyOperator(task_id="process")
task3 = EmptyOperator(task_id="end")
```

```
task1 >> task2 >> task3
```

Visual representation:

start → process → end

So, in Airflow:

DAG = Complete workflow

Task = Individual step in the workflow

Operator = Template used to define a task

Scheduler = Component that executes tasks according to the DAG

4.execute all the killer code .

docker run -d -p 8080:8080 --name airflow apache/airflow standalone

```
root@ubuntu:~$ docker run -d -p 8080:8080 --name airflow apache/airflow standalone
Unable to find image 'apache/airflow:latest' locally
latest: Pulling from apache/airflow
069feddb0f1: Pull complete
```

Docker ps

```
root@ubuntu:~$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
af83cedb1154   apache/airflow  "/usr/bin/dumb-init ..."  31 seconds ago  Up 30 seconds  0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp
root@ubuntu:~$
```

Docker logs airflow

```
try: apt install <deb name>
root@ubuntu:~# docker logs airflow
2026-06-23T05:27:24.367369Z [info] ] setup plugin alembic.autogenerate.schemas [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T05:27:24.367758Z [info] ] setup plugin alembic.autogenerate.tables [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T05:27:24.367971Z [info] ] setup plugin alembic.autogenerate.types [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T05:27:24.368122Z [info] ] setup plugin alembic.autogenerate.constraints [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T05:27:24.368285Z [info] ] setup plugin alembic.autogenerate.defaults [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T05:27:24.368364Z [info] ] setup plugin alembic.autogenerate.comments [alembic.runtime.plugins] loc=plugins.py:37
standalone | Starting Airflow Standalone
Simple auth manager | Password for user 'admin': rU7twqfmyccDvs4K
```

Simple auth manager | Password for user 'admin': rU7twqfmyccDvs4K

5.execute your first dag .

docker exec -it airflow bash

```
root@ubuntu:~$ ^C
root@ubuntu:~$ docker exec -it airflow bash
airflow@af83cedb1154:/opt/airflow$
```

ls

```
task1 ?? task2 ?? tasks
airflow@af83cedb1154:/opt/airflow/dags$ ls
first_dag.py
airflow@af83cedb1154:/opt/airflow/dags$
```

```
airflow@af83cedb1154:/opt/airflow/dags$ ls
first_dag.py
airflow@af83cedb1154:/opt/airflow/dags$ airflow dags list-import-errors
2026-06-23T05:35:54.648035Z [info] ] setup plugin alembic.autogenerate.schemas [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T05:35:54.649142Z [info] ] setup plugin alembic.autogenerate.tables [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T05:35:54.649644Z [info] ] setup plugin alembic.autogenerate.types [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T05:35:54.650335Z [info] ] setup plugin alembic.autogenerate.constraints [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T05:35:54.650813Z [info] ] setup plugin alembic.autogenerate.defaults [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T05:35:54.651400Z [info] ] setup plugin alembic.autogenerate.comments [alembic.runtime.plugins] loc=plugins.py:37
No data found
airflow@af83cedb1154:/opt/airflow/dags$ airflow dags reserialize
2026-06-23T05:38:07.033699Z [info] ] setup plugin alembic.autogenerate.schemas [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T05:38:07.034520Z [info] ] setup plugin alembic.autogenerate.tables [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T05:38:07.035604Z [info] ] setup plugin alembic.autogenerate.types [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T05:38:07.036290Z [info] ] setup plugin alembic.autogenerate.constraints [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T05:38:07.036809Z [info] ] setup plugin alembic.autogenerate.defaults [alembic.runtime.plugins] loc=plugins.py:37
2026-06-23T05:38:07.037463Z [info] ] setup plugin alembic.autogenerate.comments [alembic.runtime.plugins] loc=plugins.py:37
```

airflow dags list

```
airflow@af83cedb1154:/opt/airflow/dags$ airflow dags list
2026-06-23T05:38:39.970372Z [info] ] setup plugin alembic.autogenerate.schemas [alembic.runtime.plugins] loc=pl
2026-06-23T05:38:39.970628Z [info] ] setup plugin alembic.autogenerate.tables [alembic.runtime.plugins] loc=pl
2026-06-23T05:38:39.971241Z [info] ] setup plugin alembic.autogenerate.types [alembic.runtime.plugins] loc=plug
2026-06-23T05:38:39.971728Z [info] ] setup plugin alembic.autogenerate.constraints [alembic.runtime.plugins] lo
2026-06-23T05:38:39.972343Z [info] ] setup plugin alembic.autogenerate.defaults [alembic.runtime.plugins] loc=f
2026-06-23T05:38:39.972775Z [info] ] setup plugin alembic.autogenerate.comments [alembic.runtime.plugins] loc=f

dag_id | fileloc | owners | is_paused | bundle_name | bundle_version
-----+-----+-----+-----+-----+-----
first_training_dag | /opt/airflow/dags/first_dag.py | airflow | True | dags-folder | None
```

1 Dag

first_training_dag

training

Schedule

Latest Run

00***

1 Dag

Sort by Display Name (A-Z)

first_training_dag

training

Schedule

Latest Run

Next Run

00***

2026-06-23 11:09:44

2026-06-24 05:30:00